

Exercise 4

Exercise 4.1 Calculus of variations

Derive the Euler-Lagrange equation for a problem of the form

$$J(\mathbf{x}) = \int_{t_0}^{t_f} \phi(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \quad (1)$$

if no boundary conditions are given for the states $\mathbf{x}$ at time points $t = t_0$ and $t = t_f$.

Note: Proceed similarly to the derivation in the script for the case with fixed boundary conditions and, after partial integration, consider what additional boundary conditions are needed.

Exercise 4.2 Dynamic optimization with free terminal state

Consider the following dynamic optimization problem

$$\min_{u(\cdot)} \int_0^1 x^2 + u^2 dt \quad (2a)$$

$$\text{s.t. } \dot{x} = u - x, \quad x(0) = 1. \quad (2b)$$

Show that the optimal solution is given by

$$x^*(t) = C \exp(-\sqrt{2}t) + D \exp(\sqrt{2}t)$$

$$u^*(t) = C(1 - \sqrt{2}) \exp(-\sqrt{2}t) + D(1 + \sqrt{2}) \exp(\sqrt{2}t)$$

with the constants

$$C = \frac{(1 + \sqrt{2}) \exp(\sqrt{2})}{2(\sinh(\sqrt{2}) + \sqrt{2} \cosh(\sqrt{2}))}, \quad D = \frac{(\sqrt{2} - 1) \exp(-\sqrt{2})}{2(\sinh(\sqrt{2}) + \sqrt{2} \cosh(\sqrt{2}))}.$$

Exercise 4.3 Dynamic optimization with fixed terminal state

Consider the dynamic optimization problem

$$\min_{u(\cdot)} \int_0^1 \frac{1}{2} x^2 + \frac{a}{2} u^2 dt \quad (3a)$$

$$\text{s.t. } \dot{x} = u, \quad x(0) = 1, \quad x(1) = 0 \quad (3b)$$

with the parameter a .

- Formulate the optimality conditions for the given problem and compute the optimal trajectories.
- Plot the solution for different parameter values a with the Matlab script E4_3_plot_sol and interpret the results.
- Discuss the meaning of the adjoint states.